

Links:

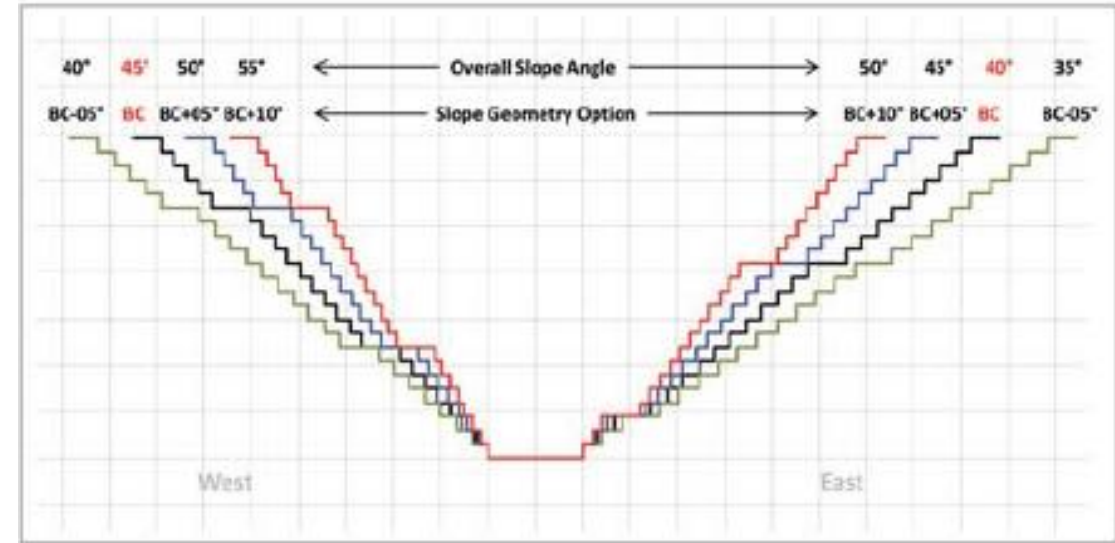
- <https://www.youtube.com/watch?v=l6nlWxzRQ8o>
- <https://www.youtube.com/shorts/6qyh61lhJd4>
- <https://www.youtube.com/watch?v=sETq5lVsY1E>



Highwall Mining:

Background:

- Due to vast development in opencast mining machinery and blasting technology, opencast mining is now planned at a greater depth. As a result, final highwall height is increased while the final highwall slope angle is reduced.
- Final highwalls in open-pit coal mines can form a starting point for other mining methods, such as highwall mining, which is basically a technique utilised after the open-pit portion of a reserve has been mined, sometimes prior to the introduction of underground mining.
- This technology enables the extraction of some thin coal seams, even less than 1.0 m, otherwise un-mineable with other existing technologies.



- Many open-pit mines in India are reaching their pit limits. Existence of surface dwellings mostly limits the expansion of currently running opencast mines.
- Sometimes the overburden is so high that coal extraction becomes uneconomical due to enhanced stripping ratios. In order to extract the locked-up coal and to extend the life of some of the open-pit mines that have reached their pit limits, highwall mining was introduced in India.
- Highwall mining involves lesser capital but can achieve good production rate and economy. Seen as a proven technology in USA, Australia and some other parts of the world.
- However, Indian coal geology is complex compared to some other parts of the world. The occurrence of thick and multiple seams, and some at close proximity, with frequently varying roof conditions etc., makes mining difficult and challenging.

Method of Mining:

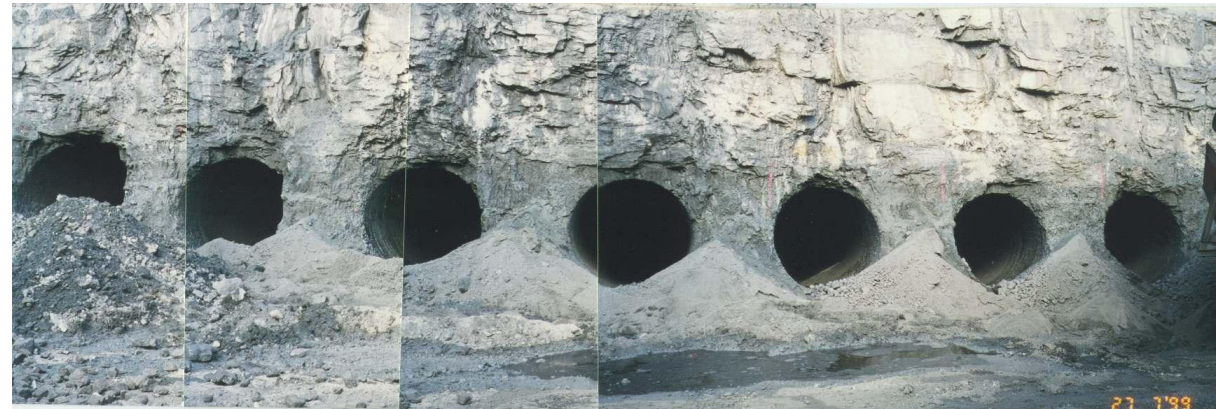
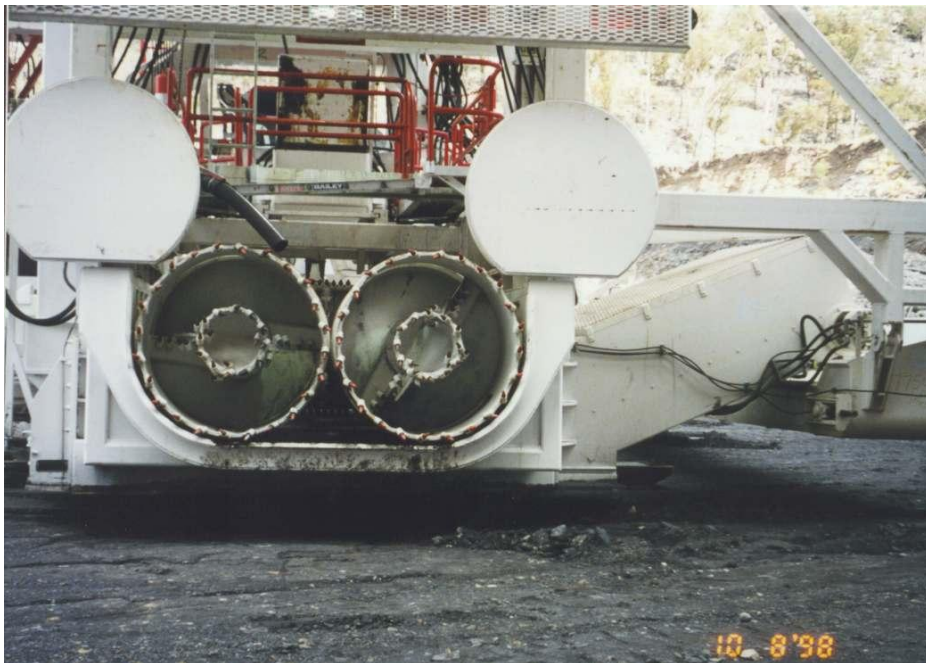
- Highwall mining operation involves driving a series of parallel entries using remotely operated continuous miners with an attached conveying system into a coal seam exposed at the highwall.
- There parallel entries (web entries) are separated by web pillars of pre-designed width, and remain unsupported, unmanned and unventilated throughout their life.



- There are two types of highwall mining systems, one being the Continuous Highwall Miner system (CHM) and the other being the auger mining system.
- CHM produces rectangular entries, usually of the typical widths 2.9 m and 3.5 m, while augur miners cut the coal seam in a circular manner with diameters of 1.35 to 1.8 m.
- Auger miners is generally restricted to a small depth of penetration (<130 m) and it could be less depending on the strength of the coal; it also has to exert more power at increased depth.
- It does not have a haulage system for a longer distance and for coal seams of varying dip and roll.

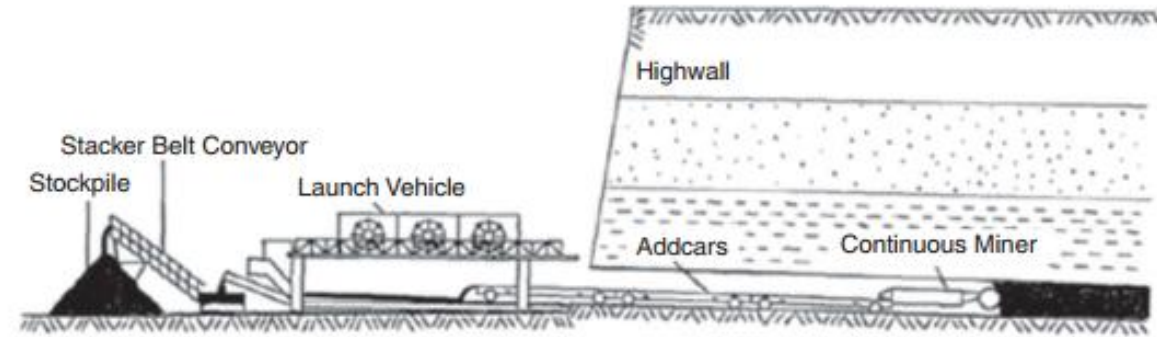


- It operates at a fixed height and is thereby restricted to one thickness in varying thickness coal seams; it is therefore not flexible and limits the coal recovery to about 30–40% in those conditions.
- It has low production rate compared to continuous miner systems such as CHM.
- Its inability to negotiate dip of the seam and diminishing power with increase in depth are other limitations.



Continuous Highwall Miner system (CHM)

- The CHM system has the following characteristics though the specifications differ from company to company:
- The system which adopts continuous miner has greater depth of penetration (about 500 m), depending on the seam dip and the conveying system adopted for conveying the debris, either traction type conveying system or push-beam type conveying system.
- Unlike the auger system, the coal fragment size is unaffected by the depth of penetration and fragment sizes depend on the lacing of picks on the drum and the power requirements.
- Since continuous miner has a cutter-head boom that can move up and down to adjust to the variable cutting heights of the coal seam, it is able to extract or recover more coal to nearly 60–80%, depending on the geological conditions.



- It can achieve higher production rate up to 0.8 to 1.0 million tonnes per year, penetrate to higher depths, and cut stronger coal.
- Depending upon the conveying traction system, the coal seams can be up-dipping (+5 deg.) and down-dipping to -15 deg.

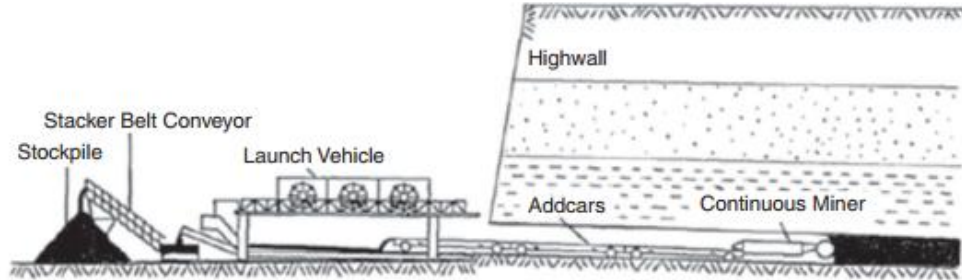


Figure 1.3 Working system of ADDCAR miner (Sasaoka et al., 2015).



Figure 1.4 Archveyor system (Sasaoka et al., 2015).

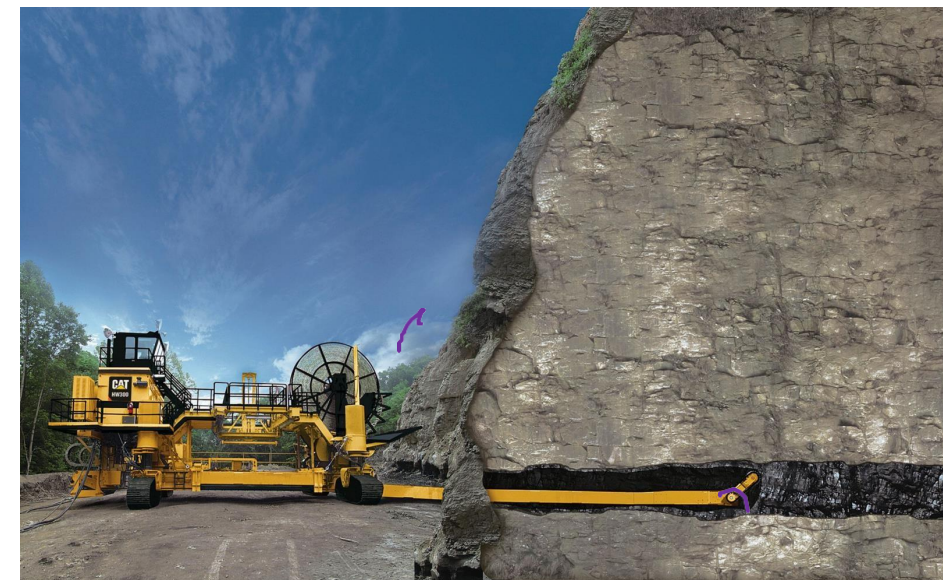
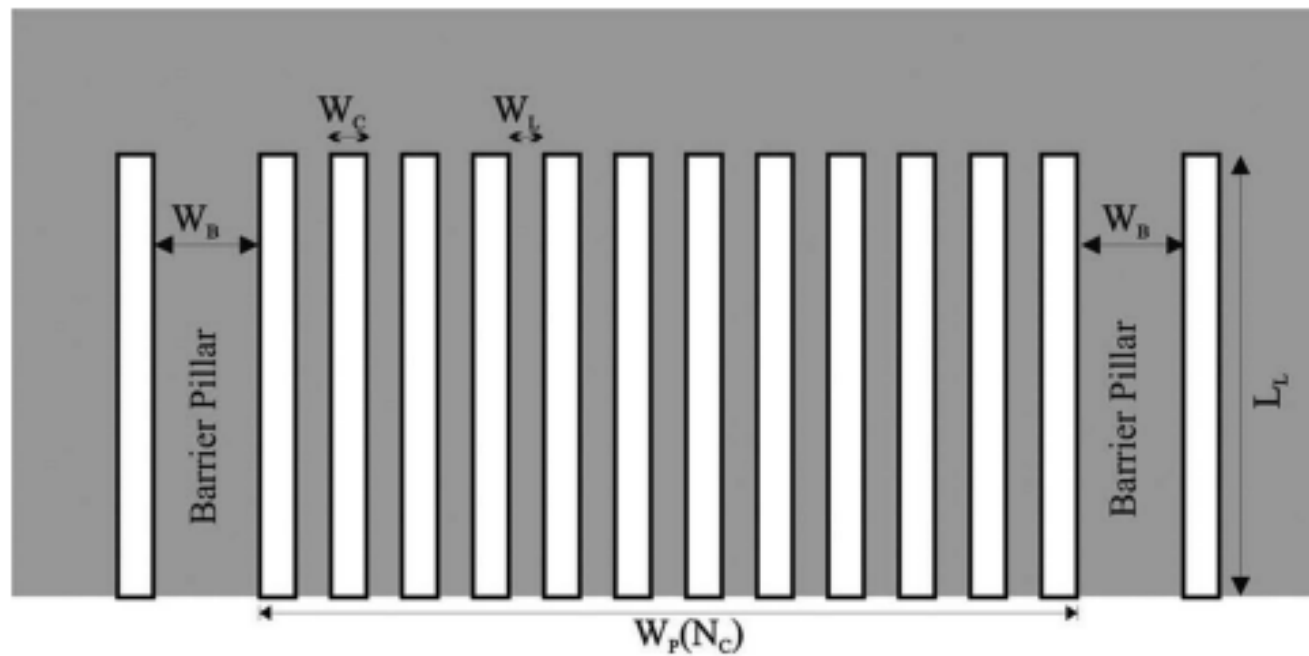


Figure 1.5 Schematic view of Caterpillar highwall miner and push beam conveyors (Hawley, 2008: www.prweb.com/releases/2008/06/prweb992884.htm).

Applicability:

Highwall mining is generally adopted in the following scenarios:

- Where open-pit reaches its pit limit and further expansion is not viable – multiple and multi-pass highwall mining is possible and has been demonstrated at the end user mine site.
- Where the roof conditions are good, moderately flat seams, competent coal seam, stable highwall and strata is free from geological disturbances and presence of stone bands in the seam.
- In thin seam conditions where open-pit/opencast/contour mining is not viable, highwall mining can be adopted with trenches – again multiple-seam highwall mining is possible by digging deeper trenches.
- In old abandoned mines, highwall mining can be adopted to extract high-value coal.
- Depending on the stability of the slope, highwall mining can be adopted in situations where transition from surface mining to underground mining is probable.
- With highwall mining, selective mining can be done of high-value coal seams, or on the blocks that are free from major faults or local geological anomalies.



Some of the benefits include:

- In highwall mining, movements of equipment can be done easily as it is mobile and is used on the surface unlike in underground mining methods where movements of equipment are limited due to space constraints.
- Highwall mining offers greater safety to men and machinery, as it is operated remotely without men involved in the highwall entries and people are not exposed to hazardous conditions such as dust, fumes, gas, roof falls, irrespirable atmosphere, etc.
- Highwall mining involves low capital and low development costs, which makes the system much cheaper compared to underground mining methods such as longwall mining.
- In some mining conditions, highwall mining provides higher coal recovery for the capital expenditure; thereby it offers higher productivity and the cost per tonne is much lower compared to underground coal mining methods for winning of coal.
- It is a clean coal technology, does not generate vast overburden waste dumps like typical open-pit mines and produces coal with practically no mix-up with waste rocks.

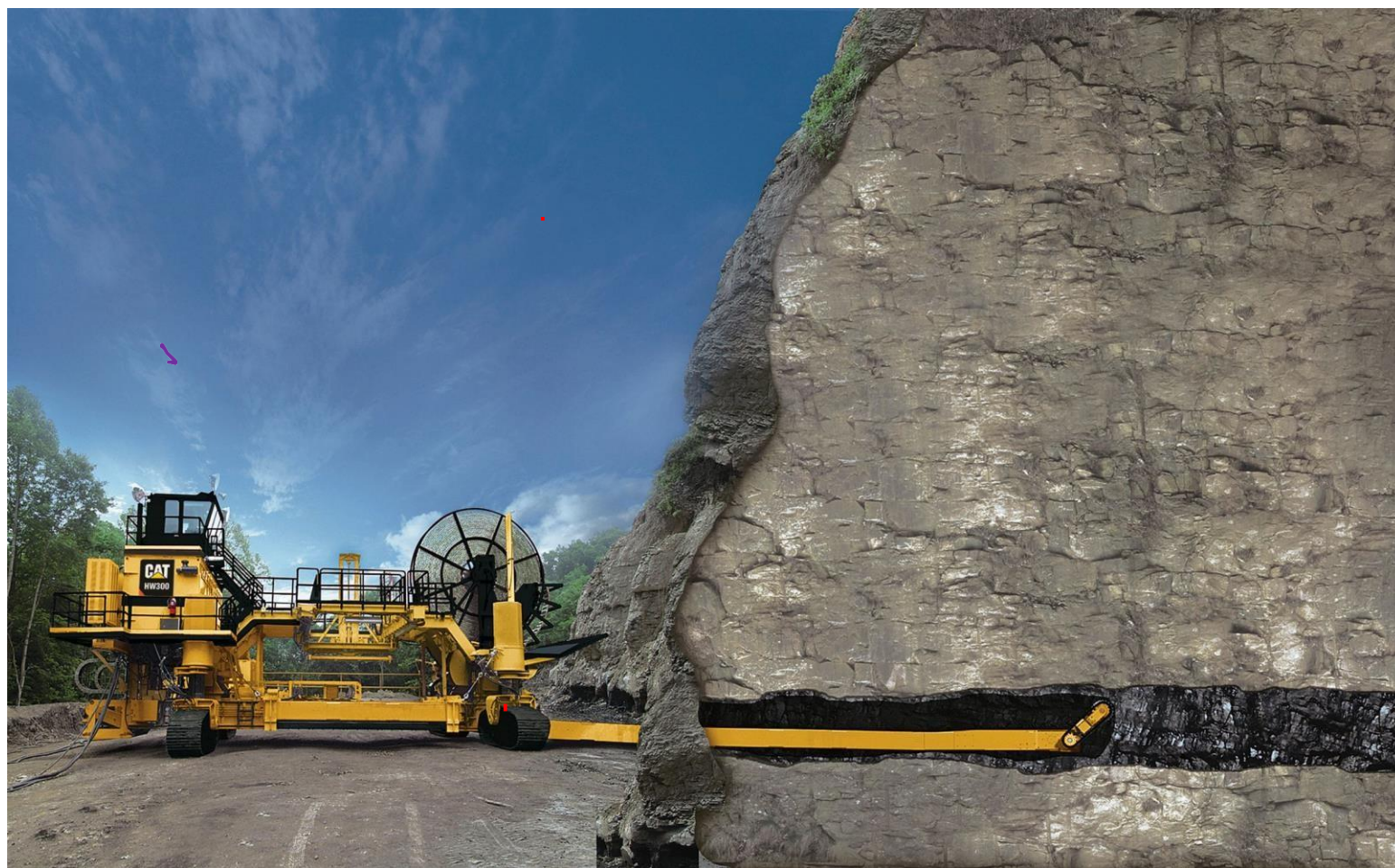
Safety:

- Accidents from the rock falls from the highwall or slope failure.
- Roof and rib falls at the portal.
- Crushing accidents caused by mobile equipment being operated in a confined space with poor visibility.
- Maintenance related injuries such as working on the launch pad, cranes lifting-related accidents, cutting and welding.
- Strain on the work force.

Highwall mining

Highwall mining equipment

- Base Unit / Advancing launcher
- Reels & chains
- Controls
- Push beams
- Cutter Unit
- Anchoring
- Right angle conveyor
- Generator.



Highwall mining

Highwall mining equipment

- **Base Unit / Advancing launcher**

- Main base unit is placed at surface where most of the drives and all controls are positioned.
- Closed operator's main control cabin is on top of the Base Unit.
- It also houses hydraulic Power-Pack to feed power to Push-beams and cutter-heads.

- **Reels & chains**

- Reel automatically unwinds and winds considering the movement of the push beam and it contains:
 - ✓ Power cable for cutter head.
 - ✓ Control cable for cutter head
 - ✓ Methane sensor cable
 - ✓ Hydraulic pressure hose for cutter
 - ✓ Cooling water hose for cutter motor



Highwall mining

Highwall mining equipment

- **Controls**

- Total operation can either be automatic through programmable logic controller, or
- Manual through touch screen technology, or
- Fully manual.
- It contains self-diagnostic health monitoring system for the machine and underground environment.

- **Push beams**

- Push beam plays an important role for parallel drivage of entries.
- contains two contra-rotating augurs for transportation of coal from the coal face.
- Articulating in-joint of two push beams provides vertical movement.



Highwall mining

Highwall mining equipment

- **Push beams**

- It allows the cutter head to negotiate the undulating contour of the coal seam.
- Normal length of push beam is about 6.0 m.

- **Cutter Unit**

- Cutter unit is basically the cutter head of continuous miner.
- Operation is fully remote controlled from the surface operator's cabin.
- Cutter modules are chosen as per thickness of coal seam.
- Power of cutter motors varies from 240KW to 350 KW.
- The entire cutter modules are interchangeable with the machine.



Highwall mining

Highwall mining equipment

- **Anchoring**

- Two machines mounted in front of the machine used for drilling into the pit.
- Two high strength pins are inserted through the base frame of the miner into the pit floor.
- During the mining cycle and retract cycle the anchoring systems stabilizes the Highwall Miner.

- **Conveyor**

- Conveyor unit facilitates loading/stacking of coal on to truck/on the ground.
- A new development at Highwall Miners is the Right Angle Discharge Conveyor.



Highwall mining

Highwall mining equipment

- **Conveyor**
 - It can be mounted to discharge coal on the right or left side of the Highwall Miner.
 - The advantage of the right angle conveyor system is the miner can work on bench of narrow width also.
- **Generator**
 - The Highwall mining system can be powered from either highline power or by utilizing the Highwall miner electric generator.
 - The mobile generator is housed in a super silenced container.
 - The mobile generator is mounted on crawlers for high maneuverability in rough terrain.



Highwall mining

Working mechanism of Push Beam Conveyors

- The coal seam is penetrated by a continuous miner propelled by a hydraulic push beam transfer mechanism.
- Cycle includes sumping (launch pushing forward) and shearing (raising and lower the cutter head boom to cut the entire height of the coal seam).
- The push beam transfer mechanism (PTM) automatically inserts a 6.01m long rectangular push beam (screw conveyor segment) into the center section of the machine between the power head and the cutter head.
- The push beam system can penetrate nearly 300m into the coal seam.
- High wall mining system use auger enclosed inside the push beam, that prevent the mined coal from being contaminated by rock debris during the conveyance process.
- Using a video imaging and/or a gamma ray sensor and/or other geo-radar system like a coal-rock interface detection sensor, the operator can see ahead projection of the seam rock interface and guide the continuous miner's progress.

Highwall mining

Mining method

- **Contour mining**
 - Highwall mining follows the contour of the coal seam, method is known as contour mining by Highwall method.
 - When any coal seam, particularly thin seam appears at incrop/outcrop, conventional method of contour mining can be done economically and safely.



Highwall mining

Mining method

- **Trench mining**
 - Entries can be made by excavating a trench to expose the coal seam and Highwall mining can be done both along the exposed coal seam available at dip and rise side.
 - This is commonly known as trench mining.

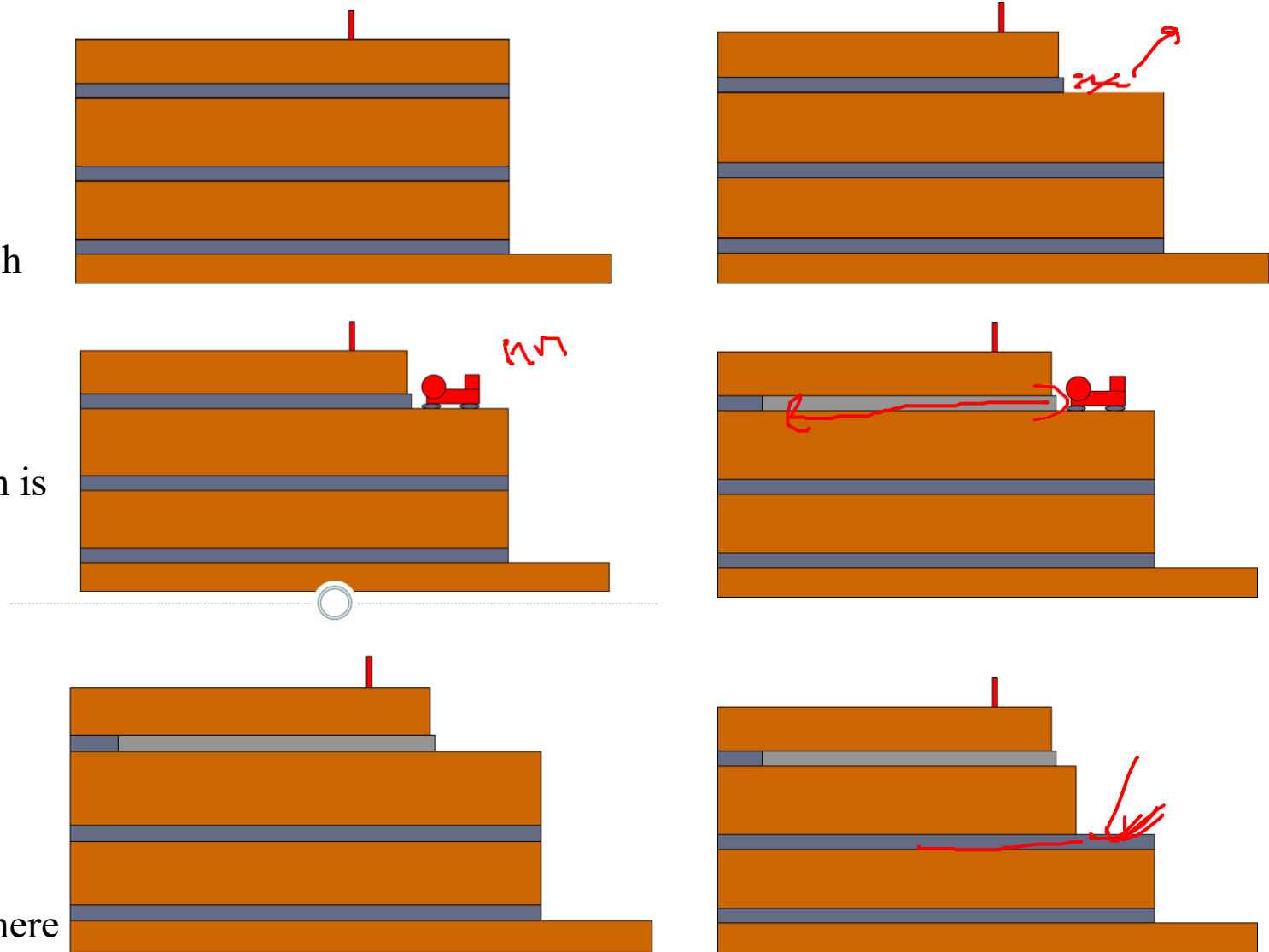


Highwall mining

Mining method

- **Bench mining**

- Mining by preparation of benches in each seam for multi-seam working is called bench mining.
- For working multi seam deposit, a bench is prepared in the top seam and mining operation is taken top downwards.
- Highwall mining can also be done by preparation of benches from bottom upwards but there is a possibility of damages to the coal of upper seams if there is any failure of web pillars at the bottom seams.

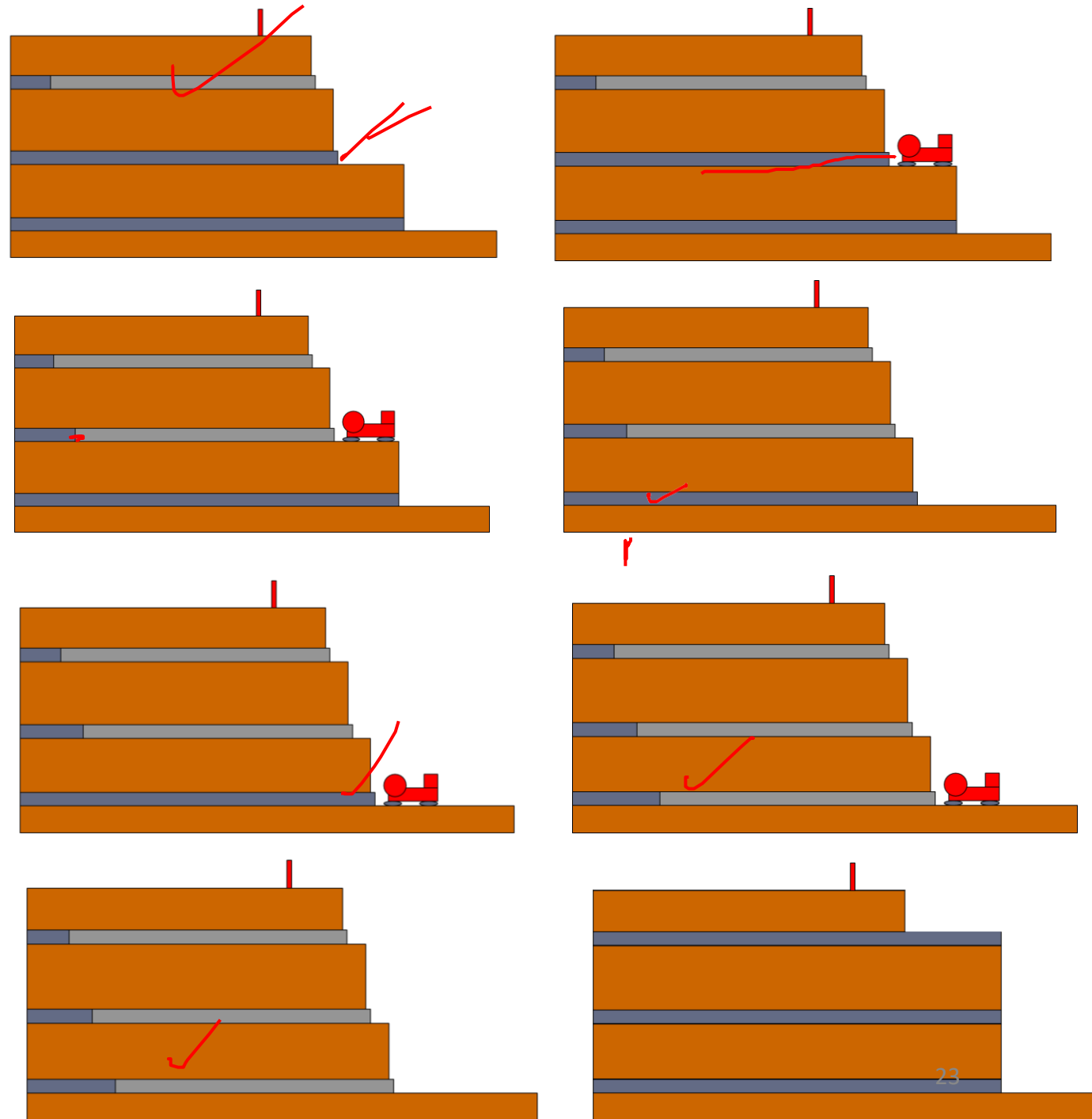


Highwall mining

Mining method

- **Bench mining**

- Mining by preparation of benches in each seam for multi-seam working is called bench mining.
- For working multi seam deposit, a bench is prepared in the top seam and mining operation is taken top downwards.
- Highwall mining can also be done by preparation of benches from bottom upwards but there is a possibility of damages to the coal of upper seams if there is any failure of web pillars at the bottom seams.



Highwall mining

Mining method

- **Highwall mining**
 - When the open pit mining has reached beyond the stripping ratio or the limit of mining operation, the coal blocked in the Highwall, can be taken out economically and safely by this method.

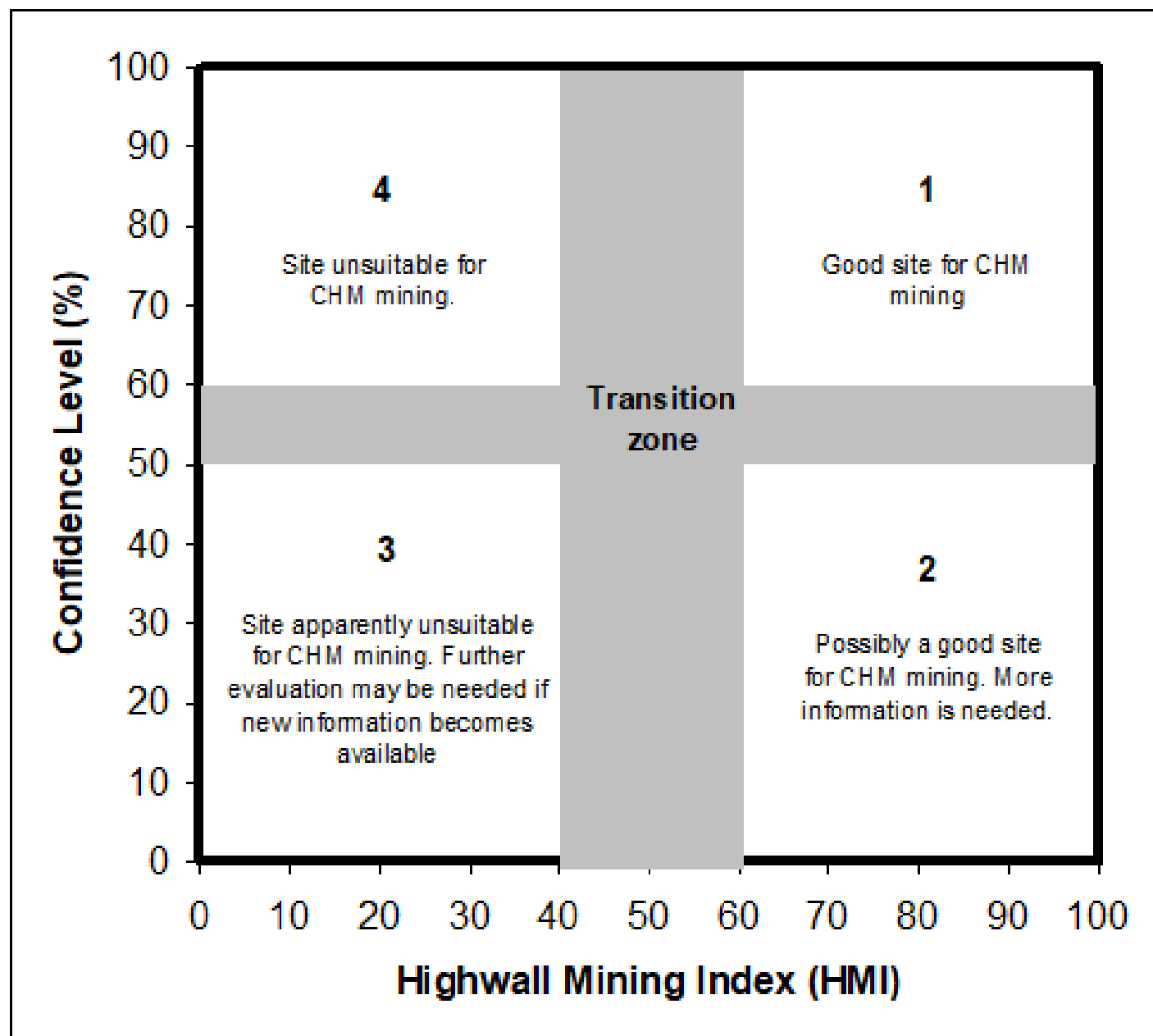


Feasibility Analysis

Table 2.1. Rating guidelines for the 15 key parameters included in the highwall mining index.

Rank	Parameter	Condition				
		Excellent (100≥rating>80)	Good (80≥rating>60)	Fair (60≥rating>40)	Poor (40≥rating>0)	No-go (rating=0)
A	Roof condition	Strong to very strong roof rocks; thickly bedded; joints with wide spacing. No alternation and weathering.	Moderately strong roof rock; medium bedded; joints with moderate spacing. Slight alternation and weathering.	Moderately weak roof rocks; <u>thinly</u> bedded; joints with close spacing. Alternation and weathering.	Weak roof rocks; very thinly bedded; joints with very close spacing. Strong alternation and weathering.	Extremely weak to very weak rock; laminated; joints with very close spacing. Severe weathering
B1	Mineable Reserves	>1500 m highwall	500 m to 1500 m highwall	250 m to 500 m highwall	100 m to 250 m highwall	<100 m highwall
B2	Seam gas (methane)	Very low methane content (<0.1m ³ /t). No concern to mining	Low methane content in general. (0.1-1m ³ /t). Higher gas only in localised pockets.	Seam methane = 1-5m ³ /t. Gas problems could be encountered occasionally in reserve.	Seam gas (5-10m ³ /t). Gassy seam and gas problems are expected to be encountered throughout.	Seam methane >10m ³ /t
B3	Seam Dip	+0° to -4°	+0° to +1°, or -4° to -7°	+1° to +2°, or -7° to -10°	+2° to +4°, or -10° to -15°	>+4°, <-15°
B4	<u>Highwall/low wall stability</u>	Smooth highwall wall. No hanging blocks. No major joint sets or faults. No risk of large scale <u>lowwall</u> instability	Localised loose rock blocks. Slight weathering. Minor joints/cracks. No discernible risk of <u>lowwall</u> instability	Common small rock falls. Dense jointing. No risk of large scale highwall instability. Very low risk of <u>lowwall</u> instability. Possible block falls.	Large rock falls. Weathered. Wedges and slabs formed by joints. Some risk of local <u>lowwall</u> instability but manageable.	Grossly unstable highwall. Significant risk of large scale <u>lowwall</u> instability.
C1	Seam continuity	Seam is consistent with no faulting or folds	Structures which slightly reduce seam thickness. Only small faults and undulations	Structures which reduce thickness by up to 50%. Seam changes direction by 5° in folds	Structures which reduce thickness over 50%. Faults and distinct folds are common.	Seam heavily affected by faulting and folding.
C2	Pit condition, preparation	No loose material, dry, no floor work.	<0.5 m loose material. Water puddles. Shave floor.	<2 m loose material. Pumping required. Some mud. Soft floor.	>2 m loose material. Major pumping and mud clean up. Floor repairs required.	Pit width <30 m.
C3	Seam quality	Medium to hard export coking coal. No in-seam parting or stone bands	Soft coking coal and export steaming coal. 1 or 2 stone bands.	Any unwashed product which is saleable. Many partings and stone bands.	Coal not in previous categories which must be washed for sale. Many stone bands.	
C4	Floor condition	Strong floor rocks. Unaffected by water.	Weak to moderately strong floor rocks. May	Weak floor rocks. Occasionally muddy.	Very weak floor rocks (clay). Wet and muddy throughout.	

Rank	Parameter	Condition				
		Excellent (100≥rating>80)	Good (80≥rating>60)	Fair (60≥rating>40)	Poor (40≥rating>0)	No-go (rating=0)
			be wet but not muddy.			
C5	Ground water	Dry environment. No water gets into the mining entries.	Seasonal dry and wet environment. Some water into entries but is manageable.	Generally wet environment. Some water problems could be expected at fault zones.	Very wet environment. Water gets into entries from the pit and from the ground. It may flood the entry.	
D1	Coal strength	Lab coal UCS > 30MPa. Limited coal joints with wide spacing.	Lab coal UCS=20-30MPa. Coal joints with moderate spacing.	Lab coal UCS=10-20MPa. Coal joints with close spacing.	Lab coal UCS <10MPa. Heavily cleated. Coal joints with very close spacing.	
D2	Cover depth (overburden + spoil)	Overburden depth <80 m in the reserve	Overburden depth 50 – 100 m.	Overburden depth 60 – 120 m, possible stress concentration near highwall	Overburden depth 80 – 150 m, possible significant stress concentration near highwall	
D3	Subsidence	Surface subsidence not an issue.	Minor subsidence effects but all contained on lease. Minimal rehabilitation efforts.	Some subsidence or water effects off site but controllable. Rehabilitation could be costly.	No subsidence is allowed. Significant water effects off site. Major rehabilitation costs. Subsidence control is critical.	
D4	Seam thickness	>3.5 m	2.5 m- 3.5 m	2.0 m – 2.5 m	1.5 m - 2.0 m	<1.5 m.
D5	Mobilisation and service	Equipment, service capacity and skilled mining crew are available on site.	Equipment, service capacity and skilled mining crew are available in mines nearby.	Equipment, service capacity and skilled mining crew are available interstate.	Equipment and service capacity are available overseas. New mining crews need to be trained.	



Geological Characterization of Highwall Mining Operations



